INCH-POUND

MIL-R-39016/40F
4 June 1992
SUPERSEDING
MIL-R-39016/40E
29 January 1991

MILITARY SPECIFICATION SHEET

# (F) RELAYS, ELECTROMAGNETIC, 4PDT, LOW LEVEL TO 2.0 AMPERES (0.100-INCH TERMINAL SPACING)

(F) Inactive for new design after 29 January 1991.
For new designs refer to MIL-R-39016/14.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-R-39016.

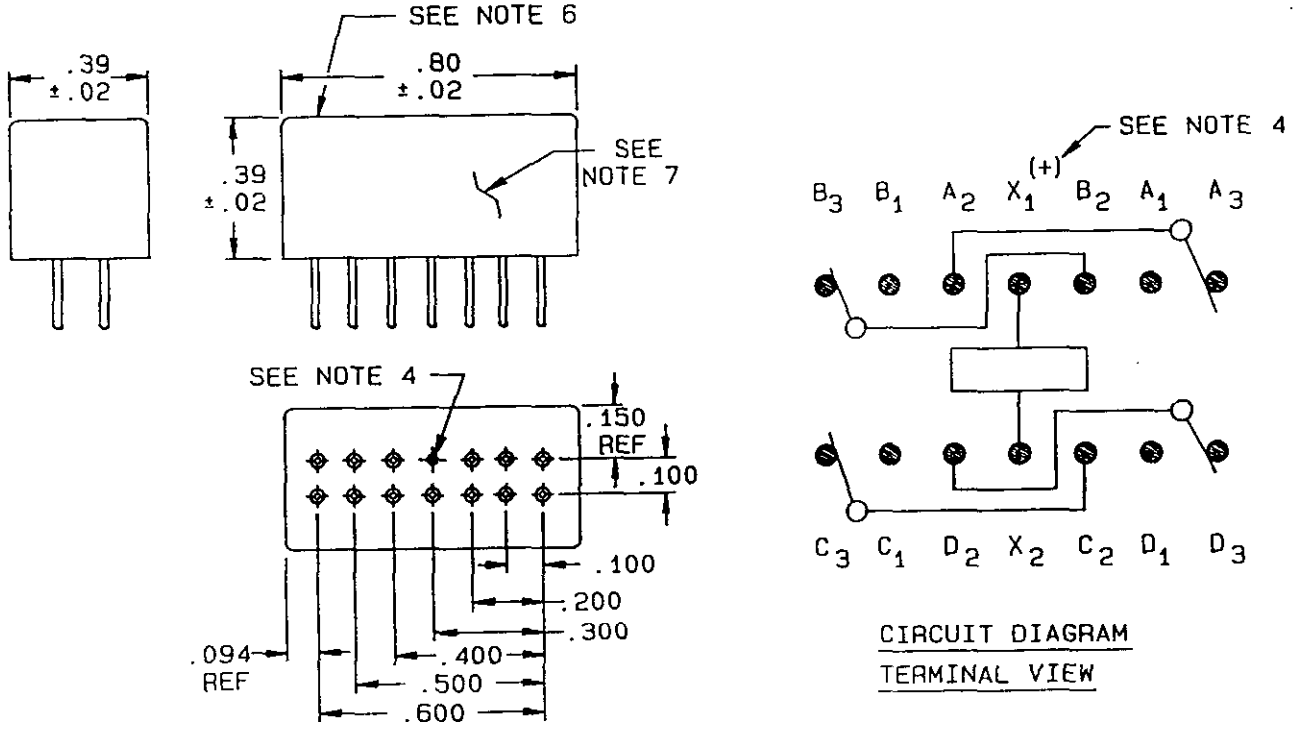



FIGURE 1. Dimensions and configurations.

(F) denotes changes


DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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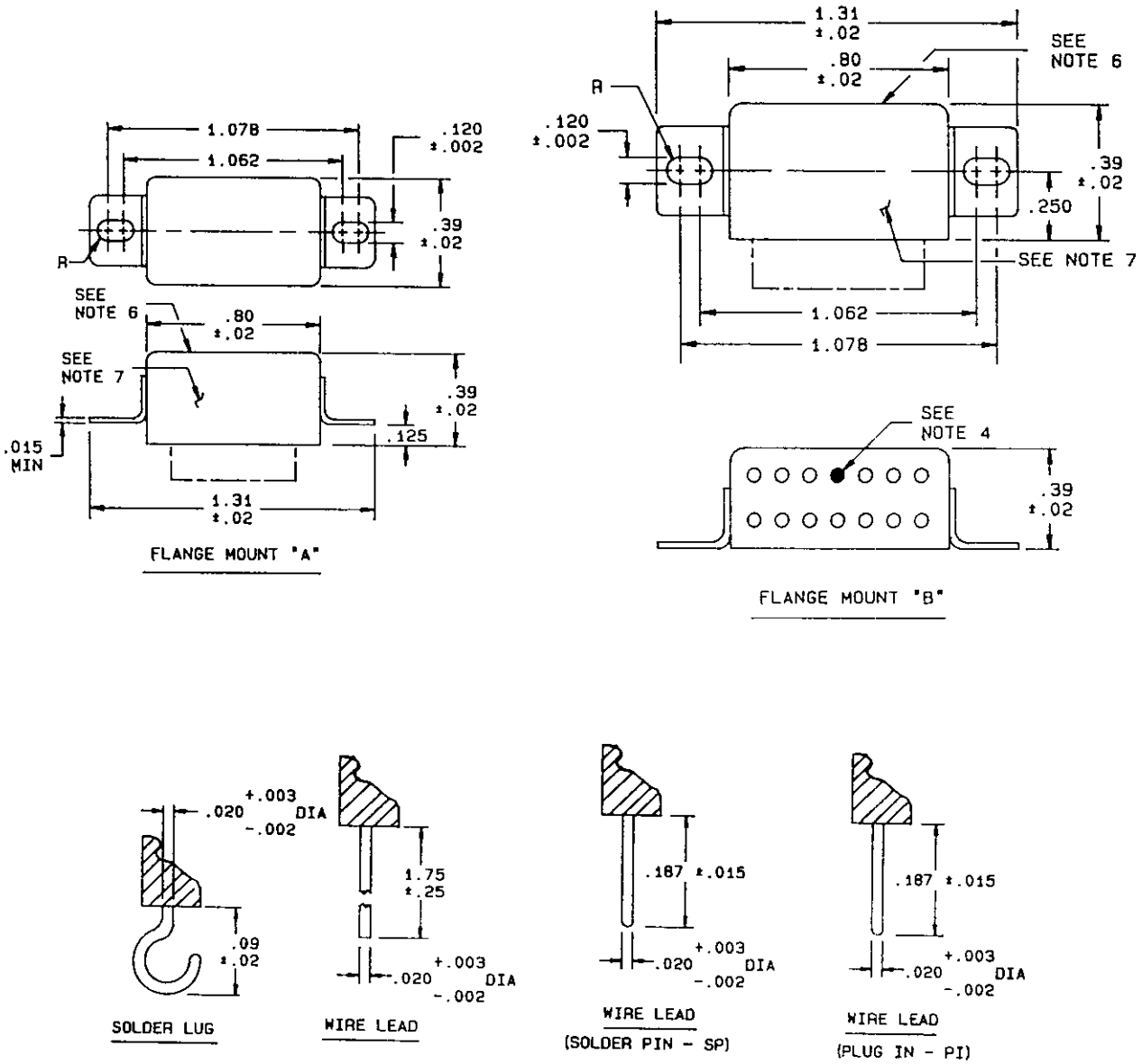



FIGURE 1. Dimensions and configurations - Continued.



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| Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|
| .002 | 0.05 | .120 | 3.05 | .500 | 12.70 |
| .003 | 0.08 | .125 | 3.18 | .600 | 15.24 |
| .005 | 0.13 | .150 | 3.81 | .800 | 20.32 |
| .015 | 0.38 | .187 | 4.75 | 1.062 | 26.97 |
| .020 | 0.51 | .200 | 5.08 | 1.078 | 27.38 |
| .090 | 2.29 | .250 | 6.35 | 1.310 | 33.27 |
| .094 | 2.39 | .300 | 7.62 | 1.750 | 44.45 |
| .100 | 2.54 | .390 | 9.91 | | |
| .110 | 2.79 | .400 | 10.16 | | |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. X1 terminal shall be identified with a contrasting bead. Relays which require coil polarity observance shall have a plus (+) sign placed on the schematic diagram as shown.
5. Shape of lug (solder) terminals is optional.
6. Schematic diagram shall be marked on this surface.
7. Identification marking shall be marked on this surface.
8. Terminal numbers in circuit diagram are for reference only. Numbers shall not appear on relay.
9. Wire lead, plug-in (PI), shall be gold-plated in accordance with MIL-G-45204, type II, class I, nickel underplating, 50 to 150 microinches.

FIGURE 1. Dimensions and configurations - Continued.



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REQUIREMENTS:

CONTACT DATA:

Load ratings:

High level (relay case grounded):

Resistive: 2 amperes at 28 V dc maximum; 0.3 ampere at 115 V ac, (60 and 400 Hz). 400 Hz life test not required for qualification testing.

Inductive: 0.75 ampere maximum at 28 V dc maximum with 200 millihenries minimum.

Lamp: 0.200 ampere at 28 V dc.

Low level: 10 to 50 μA at 10 to 50 mV dc or peak ac.

Intermediate current: Applicable.

Contact resistance and voltage drop:

Initial: 0.050 ohm maximum.

High level:

During life: Not more than 5 percent of open circuit voltage.

After life: 0.100 ohm maximum.

Low level:

During life: 50 ohms maximum.

After life: 0.150 ohm maximum.

Intermediate current:

During intermediate current: 3 ohms maximum.

After intermediate current: .100 ohm maximum.

(F) Contact bounce: 2.0 milliseconds (ms) maximum.

(F) Contact stabilization time: 2.5 milliseconds (ms) maximum.

Overload (high level only): Two times rated current.

COIL DATA: (See table I.)

Operate time: 5 ms maximum over temperature range with rated coil voltage.

Release time: 5 ms maximum over temperature range from rated coil voltage.



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ELECTRICAL DATA:

Insulation resistance: 1,000 megohms minimum, except the resistance between coil and case at high temperature shall be 500 megohms, or greater.

Dielectric withstanding voltage:

| | Sea level V rms (60 Hz) | Altitude V rms (60 Hz) |
|---|---|---|
| Between case, frame, or enclosure, and between all contacts in the energized and deenergized positions | 500 | |
| Between case, frame, or enclosure and coil(s) | 500 | 350 |
| Between all contacts and coil(s) | 500 | All terminals to case |
| Between open contacts in the energized and deenergized positions | 350 | |
| Between contact poles | 500 | |

ENVIRONMENTAL DATA:

Temperature range: -65°C to +125°C.

Vibration (sinusoidal): MIL-STD-202, method 204 (20 g, 10 to 2,000 Hz). Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum closure for open contacts.

Vibration (random): MIL-STD-202, method 214, test condition IG. Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum closure for open contacts (applicable to qualification and group C testing only).

Shock (specified pulse): MIL-STD-202, method 213, test condition C (100 g). Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum closure for open contacts.

Magnetic interference: Applicable.

Resistance to soldering heat: Applicable, except for plug-in relays.

Acceleration: 100 g.



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PHYSICAL DATA:

Termination: See figure 1 and table I.

Solderability: Applicable, except for plug-in terminals.

Terminal strength: Applicable (1.0 ±0.3 pounds pull).

Dimensions and configuration: See figure 1.

Weight: 0.35 ounce maximum.

Identification marking (full): Applicable, except marking surface for date code is optional.

LIFE TEST REQUIREMENTS:

(F) Intermediate current: 50,000 cycles.

(F) Electrical: 100,000 cycles.

(F) Mechanical: 1,000,000 cycles.

(F) Part or Identifying Number (PIN): M39016/40- (plus dash number from table I).



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(F) TABLE I. Dash numbers and characteristics. 1/

| Dash number | | | | Mounting | Coil voltage (V dc) | | Coil resistance (±10%) (+25°C) Ohms | Specified pickup value (voltage) (V dc) (over temp. range) | Specified hold value (voltage) (V dc) (over temp. range) | Specified dropout value (voltage) (V dc) (over temp.) range) |
|---|---|---|---|---|---|---|---|---|---|---|
| Terminals | | | | | | | | | | |
| Solder lug | Wire lead (SP) 2/ | Wire lead | Wire lead (PI) | | Rated 3/ | Max | | | | |
| -001 | -007 | -013 | -019 | Flange A | 26.5 | 32.0 | 390 | 18.0 | 14.0 | 1.2 |
| -002 | -025 | -014 | --- | Flange B | | | | | | |
| --- | -008 | -028 | -020 | No mount | | | | | | |
| -003 | -009 | -015 | -021 | Flange A | 12.0 | 15.0 | 100 | 9.0 | 5.8 | 0.6 |
| -004 | -026 | -016 | --- | Flange B | | | | | | |
| --- | -010 | -029 | -022 | No mount | | | | | | |
| -005 | -011 | -017 | -023 | Flange A | 6.0 | 8.0 | 25 | 4.5 | 2.9 | 0.3 |
| -006 | -027 | -018 | --- | Flange B | | | | | | |
| --- | -012 | -030 | -024 | No mount | | | | | | |

1/ Each relay possesses high level and low level capabilities. However, relays previously tested or used above 10 mA resistive at 6 V dc maximum or peak ac open circuit are not recommended for subsequent use in low level applications.

2/ CAUTION: Solder pin relays are not intended for plug-in socket use.

3/ CAUTION: The use of any coil voltage less than the rated voltage will compromise the operation of the relay.



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QUALIFICATION INSPECTION:

Qualification inspection and sample size: See table II.

(F) TABLE II. Qualification inspection and sample size.

| Single submission | Group submission | |
|---|---|---|
| 50 units plus 1 open unit. One failure allowed. | M39016/40-001 | 50 units plus 1 open unit. One failure allowed. |
| Qualification inspection as applicable. | | Qualification inspection as applicable. |
| | M39016/40-009<br>M39016/40-024 | 2 units each PIN, qualification inspection table, group II. No failures allowed. |
| | M39016/40-014 | 2 units, qualification inspection table, group II, and shock, vibration, acceleration, terminal strength, and seal. No failures allowed. |

SUPERSESSION DATA: See table III.

(F) TABLE III. Supersession data.

| Superseded PIN M5757/90- | New PIN M39016/40- | Superseded PIN M5757/90- | New PIN M39016/40- |
|---|---|---|---|
| 001 and 002 | 001 | 025 and 026 | 013 |
| 003 and 004 | 002 | 027 and 028 | 014 |
| 005 and 006 | 003 | 029 and 030 | 015 |
| 007 and 008 | 004 | 031 and 032 | 016 |
| 009 and 010 | 005 | 033 and 034 | 017 |
| 011 and 012 | 006 | 035 and 036 | 018 |
| 013 and 014 | 025 | 037 and 038 | 007 |
| 015 and 016 | 008 | 039 and 040 | 009 |
| 017 and 018 | 026 | 041 and 042 | 011 |
| 019 and 020 | 010 | 043 and 044 | 028 |
| 021 and 022 | 027 | 045 and 046 | 029 |
| 023 and 024 | 012 | 047 and 048 | 030 |

(F) QUALITY ASSURANCE PROVISIONS:

Group B not required.
Groups A and C required. Group C testing shall be performed on normal production relays. Lots shall not be generated solely for group C testing, unless requested by the qualifying activity.



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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AT, ME
Navy - AS, OS
DLA - ES

User activity:
Army - AV
Navy - SH

Preparing activity:
Navy - EC

Agent:
DLA - ES

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